WHY THREE BLOCKS? THE LIMITED AUGMENTED ZARANKIEWICZ NUMBER IN FIXED WIDTH

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Abstract. The limited augmented Zarankiewicz number $z_L(m,n)$ of Qi, Cui and Xu bounds the maximum sum-of-squares rank of $m \times n$ biquadratic forms from below: it counts the cells of a maximum C_4-free set in the $m \times n$ grid together with pairs of further cells that avoid two forbidden patterns. We determine $z_L(m,3) = \lfloor 3m/2 \rfloor + 2$ for every $m \geq 3$, answering a question of Qi, Löfberg and Chen, with a proof formalized in Lean; it also shows that the gap to their weak variant grows like $m/6$. For a fixed number n of columns we prove that each new row adds ρ_n pairs up to a bounded error, where ρ_n is a rational number given by finitely many linear programs, and that the error is eventually periodic. Pairs placed around a hollow triangle of three column blocks give $\rho_n \geq \lambda_n = n/3 - 1/2 + O(1/n)$; balanced blocks are optimal within this family, and $\rho_n = \lambda_n$ for $n \leq 6$ by rational dual certificates checked in exact arithmetic; we conjecture equality for all n. Consequently, in every fixed width the original criterion stays linearly below the recursive-line, signed and second-order ones. We also find the slope $\lfloor (n-1)^2/4 \rfloor/(n-1)$ of Lebedev's canonical base for every n, the exact value of $z_L(m,4)$ for every $m \geq 6$ by certified computation, fourteen new exact values by exhaustive computer search, and a three-dimensional analogue in which the naive bound of one quarter fails already in $3 \times 3 \times 5$, the smallest such box. The admissibility lemmas behind the constructions are formalized in Isabelle/HOL.

1. A question about zeros and ones

How many ones can an $m \times n$ matrix of zeros and ones contain without four of them at the corners of a rectangle? In 1951 Zarankiewicz asked, for square matrices, how many ones force a $k \times k$ submatrix of ones [22]; the matrix is one "consisting exclusively of 0's and 1's", as Kővári, Sós and Turán put it when they proved the upper bound of order $n^{3/2}$ for 2×2 three years later [14]. Reiman sharpened the bound and observed that finite projective planes attain it [20], and Guy computed tables of exact values [12, 13]; the history of these *degenerate* extremal problems is told by Füredi and Simonovits [9]. When one side is long compared with the other the problem becomes exactly solvable; a classical formula of this kind is due to Čulík [7] (as quoted in [2]), and recent work of Chen, Horsley and Mammoliti determines the values near the triple-system threshold [2, 3]. For this note one regime matters. Write $z(m,n)$ for the maximum number of ones; when $m \geq \binom{n}{2}$ it is

$$z(m,n) = m + \binom{n}{2},$$

attained by one row for each pair of columns and single ones in all other rows; every extremal matrix has this form.

Indeed two rows share at most one column, so the $\binom{d_r}{2}$ pairs of columns covered by the rows are distinct, $\sum_r \binom{d_r}{2} \leq \binom{n}{2}$, and $d \leq 1 + \binom{d}{2}$ for every integer $d \geq 0$, with equality only for $d = 1, 2$.

Seventy-five years after Zarankiewicz, the same matrices returned from an unexpected side. In 2026 Qi, Cui and Xu noticed that a C_4-free matrix is a certificate in real algebraic geometry: it forces many squares in a sum of squares of bilinear forms (Section 2). To push the certificate further they let the matrix carry *pairs* of extra cells, and asked how many pairs fit. That is the number $z_L(m,n)$ of this paper. Within a few months a series of papers had computed small values and bounds, and Lebedev had found the density one third when both sides grow (Section 3).

This paper follows one question through to its answer: *if the number of columns is fixed and rows keep coming, how many pairs does each row buy?* For three columns the answer is exactly one half (Theorem A, with a Lean proof). In general a pair must be paid for with empty cells, and the cheapest payment is arranged around a triangle of three blocks of columns whose face stays empty (Section 4). This gives an explicit slope λ_n for every n (Theorem B); certified linear programs show that nothing does better for $n \le 6$, and the same programs prove that the slope always exists and is rational (Theorem C, Section 6). An exact finite description by profiles of triples then settles four columns for every $m \ge 6$ (Theorem E) and shows that the error term is eventually periodic in every width (Theorem F); for Lebedev's canonical base the slope is found for every n (Theorem D). Exact values and a three-dimensional variant close the paper. Figure 1 shows how the statements depend on each other and what each one rests on.

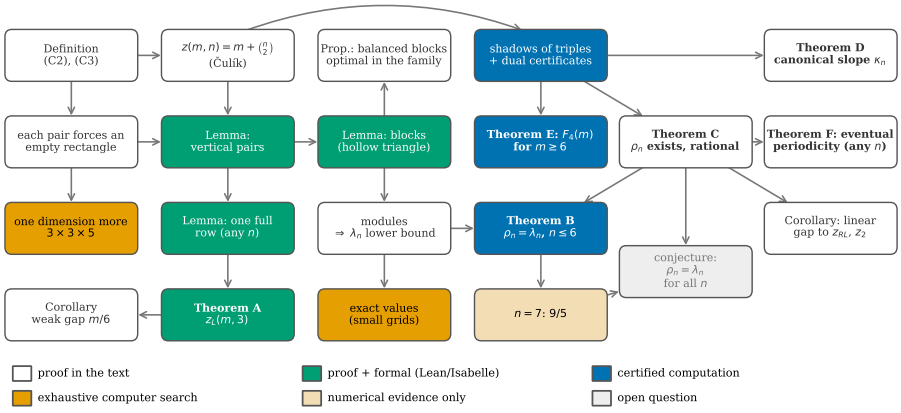


FIGURE 1. A map of the paper. Each box is a statement and an arrow means "is used in". The colour says what the statement rests on: a proof in the text; a proof also formalized (Theorem A in Lean; the vertical-pair, full-row and blocks lemmas in Isabelle/HOL, see Formalizations at the end); a computation certified in exact arithmetic; an exhaustive computer search; numerical evidence only; or an open question. Theorems A–F are in bold.

2. From cells to squares

Why should a combinatorialist care about sums of squares? A biquadratic form in $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ that is nonnegative need not be a sum of squares of bilinear forms: Calderón showed that it is when $m = 2$ or $n = 2$ [1], and Choi gave a 3×3 form that is not [5]; see [6] for the wider picture. Among forms that *are* sums of squares one asks for the least number of squares needed, and for its maximum $\mathrm{BSR}(m,n)$ over all such forms, of which only small cases and bounds are known [25]. Write a cell (i,j) as the bilinear monomial $x_i y_j$. A rectangle of four cells must be avoided, because its four squares are only two: $x_i^2y_j^2 + x_i^2y_q^2 + x_p^2y_j^2 + x_p^2y_q^2 = (x_iy_j + x_py_q)^2 + (x_iy_q - x_py_j)^2$. Qi, Cui and Xu [25] show that a C_4-free set of cells gives a sum of squares $\sum x_i^2y_j^2$ that cannot be written with fewer squares, and that one can do better: a square $(x_iy_j + x_py_q)^2$ uses two further cells at once, and under two local conditions the enlarged sum is still irreducible. This gives the chain

$$\mathrm{BSR}(m,n) \;\ge\; z_A(m,n) \;\ge\; z_L(m,n) \;\ge\; z(m,n)$$

of lower bounds for the maximum sum-of-squares rank of biquadratic forms; z_A is defined like z_L below but without asking the base E_1 to be maximum.

Definition 2.1 ([26, Def. 2.1])**.** Let E_1 be a C_4-free set of cells with $|E_1| = z(m,n)$, and let E_2 be a family of unordered pairs of distinct cells outside E_1, no cell in two pairs; put $O = E_1 \cup \bigcup E_2$. The family is *admissible* if every pair $\{(i,j),(p,q)\}$ satisfies

(C2) if $i \neq p$ and $j \neq q$, the opposite cells $(i,q),(p,j)$ are not both in O;
(C3) no cell (k,ℓ) with $k \notin \{i,p\}$, $\ell \notin \{j,q\}$ has $(k,\ell),(k,j),(k,q),(i,\ell),(p,\ell) \in O$.

The *limited augmented Zarankiewicz number* $z_L(m,n)$ is the maximum of $|E_1| + |E_2|$.

What do the two conditions say? Condition (C3) has a clean geometric meaning, which drives everything below: for a pair, let B be the set of rows other than i,p occupied in both columns j,q, and C the set of columns other than j,q occupied in both rows i,p; then

(C3) $\iff$ the rectangle $B \times C$ is completely empty.

So every pair forces an empty rectangle, and (C2) forces an empty corner. Pairs are paid for with holes; the question is how cheaply.

Both conditions are one condition on distances. Let R_O be the graph on the cells of O in which two cells are adjacent when they are opposite corners of a rectangle whose four corners lie in O. A pair at distance 1 is exactly a violation of (C2), and a common neighbour (k,ℓ) of (i,j) and (p,q) is exactly a witness for (C3): the two rectangles $\{i,k\} \times \{j,\ell\}$ and $\{p,k\} \times \{q,\ell\}$ are the five cells of (C3) plus the pair. Hence

a family E_2 is admissible $\iff$ the two cells of every pair are at distance at least 3 in R_O.

This is the form that generalizes to higher dimensions in Section 8.

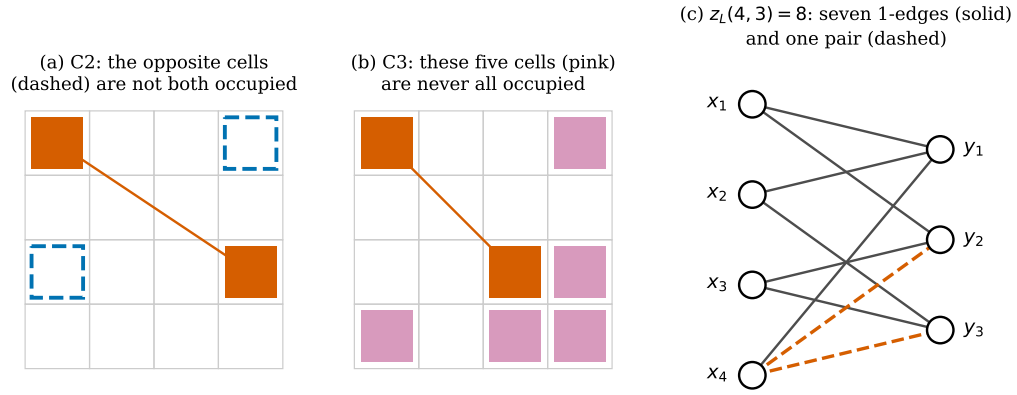


FIGURE 2. The definition at a glance. (a) Condition (C2): for the pair (orange), the two opposite cells (dashed) are not both occupied. (b) Condition (C3): with the pair in orange, the five pink cells, lying in one further row k and one further column ℓ, are never all occupied. (c) The case 4×3 as a bipartite graph, as in [25]: rows x_i, columns y_j; the seven cells of a maximum base (rows $\{1,2\},\{1,3\},\{2,3\},\{1\}$) as solid edges and the single pair, which fills row 4, as two dashed edges; so $z_L(4,3) = 8$.

Why does exactly one pair fit in 4×3*?* The base of Figure 2(c) has rows $\{1,2\},\{1,3\},\{2,3\}$ and $\{1\}$. The pair fills the last row completely; it is admissible because no other row is full, so no witness for (C3) exists. A second pair would need a second full row, and two full rows always produce a witness: this is the whole idea of the three-column theorem below.

3. The question in fixed width

The series moved fast. The first paper (March 2026) introduced z_A and z_L and computed all values with $m, n \le 4$ [25]; the second (April) built lower bounds from complete graphs and settled 5×3, 5×4 and 6×4 [26]; the third (May) turned the problem into integer programs, with a table up to 7×7 and $z_L(10, 5) = 26$ [29]; a Zenodo supplement of Howard-Stanley documents $z_L(6, 6) = 20$ [23]; Liu and Song extended the complete-graph constructions to blocks K_{5t} [17]. A weak variant with acyclic dependencies followed in July [27], and its $m \times 3$ paper computed the original $z_L(m, 3)$ for $m \le 13$ by exhaustive search and posed the value for all m as an open problem [28, §9]. In parallel, Löfberg and Qi studied a second-order number z_2 and found exact values in three and four columns [18]: fixed width is where exact answers appear first. When both dimensions grow, Lebedev [24] proved that, along sequences with $n \to \infty$ and $m/n^2 \to c \in (0, \infty)$, the density $z_L(m, n)/(mn)$ tends to $1/3$ if and only if $c \ge 1/12$ [24, Thms. 14–15], with a stability theorem: near-optimal configurations look like three diagonal blocks left empty [24, Thm. 5]. His lower bounds come from a construction with three blocks of rows and columns and reciprocal pairs, first with an empty base and then with a maximum one [24, Thms. 4 and 13]. For a *fixed* number of columns these results leave the linear term in m open.

If n is fixed and m grows, how many pairs does each new row buy? Put $F_n(m) = z_L(m, n) - z(m, n)$.

Theorem A. *For every $m \ge 3$,*

$$z_L(m, 3) = m + 3 + \left\lfloor \frac{m-2}{2} \right\rfloor,$$

and every maximum base admits at most $\lfloor (m-2)/2 \rfloor$ pairs. The proof is formalized in Lean 4.

Theorem B. *For every $n \ge 3$, $\liminf_{m\to\infty} F_n(m)/m \ge \lambda_n$, where*

$$\lambda_n = \begin{cases} \dfrac{2n-3}{6}, & n \equiv 0 \pmod 3, \\ \dfrac{(n-1)(2n-5)}{6(n-2)}, & n \equiv 1 \pmod 3, \\ \dfrac{(n-2)(2n-1)}{6(n-1)}, & n \equiv 2 \pmod 3, \end{cases}$$

and $\lim_{m\to\infty} F_n(m)/m = \lambda_n$ for $3 \le n \le 6$: the slopes are $1/2$, $3/4$, $9/8$ and $3/2$.

In Section 6 we also show that the slope exists for every n: it is a rational number $\rho_n \ge \lambda_n$, the optimum of a finite family of linear programs, and $F_n(m) = \rho_n m + O_n(1)$ (Theorem C).

4. Why three blocks?

What is the cheapest way to pay for pairs with holes? Project the columns onto a few blocks and let every row occupy the union of two blocks. A pair joins an endpoint of a row of type $\{A, C\}$ lying in block A with an endpoint of a row of type $\{B, C\}$ lying in block B. Both opposite cells fall in blocks these rows avoid, and a witness for (C3) would have to meet all three blocks, which no row does (Figure 3). Geometrically the rows are the sides of a triangle whose face is never filled, and each pair is an angle around that hollow face.

The number three has a long history here. In 1941 Turán determined the largest graph on N vertices with no complete subgraph K_{r+1}: the complete r-partite graph with equal parts [21], the result from which extremal graph theory grew [9]. It reappears as follows. Take a square matrix with empty base and empty diagonal, let the occupied set be the adjacency matrix of a graph G, and pair each edge with its transpose, $\{(i, j), (j, i)\}$. Then (C2) holds automatically, since the opposite cells lie on the diagonal, and a witness for (C3) is exactly a K_4 through the edge ij. So the best such configuration is Turán's

complete tripartite graph, and the three blocks are Turán's three parts. (Symmetry alone is not enough: the four-cycle is symmetric and K_4-free, but the pair $\{(1,2),(3,4)\}$ breaks (C2).) The lemma below is a formulation of Lebedev's construction [24, Thms. 4, 13] in the form that fixed width requires: column blocks only, of any sizes, and rows of mixed types.

Lemma 4.1 (blocks)**.** *Partition the columns into blocks, give every row of the grid a type T of at most two blocks with its occupied support inside $\bigcup_{x\in T} B_x$, and let every pair join an endpoint of a row of type T in a block $x \in T \setminus T'$ to an endpoint of a row of type T' in a block $y \in T' \setminus T$. Then the pairs are admissible.*

Proof. The opposite cells lie in block y of the first row and block x of the second, which these rows avoid: (C2). A column common to both rows lies in a block $z \in T \cap T'$ distinct from x, y, so a (C3) witness row would meet three blocks. □

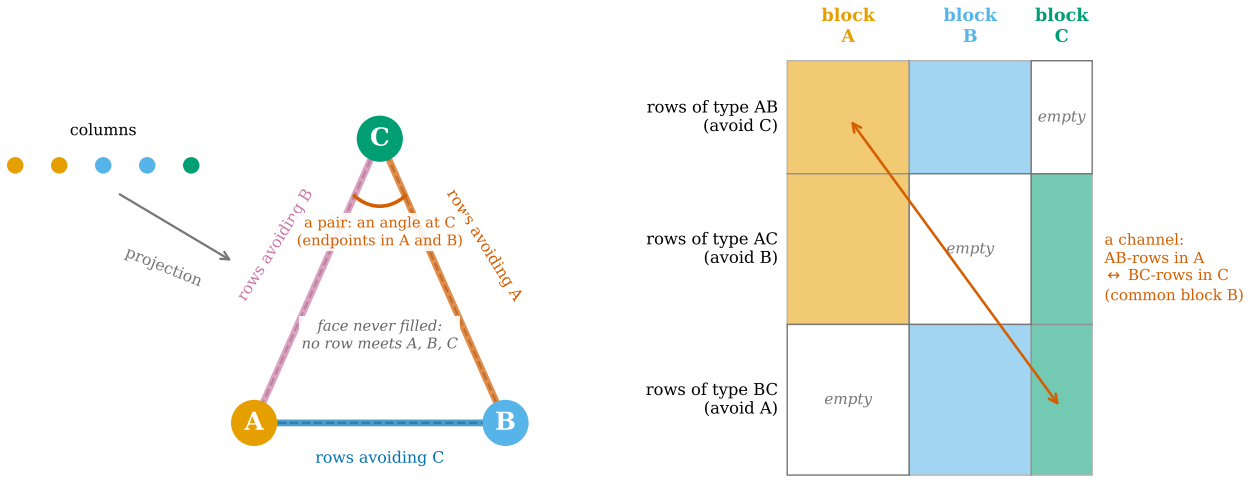


FIGURE 3. The mechanism behind the three-block constructions. Left: the columns are projected onto three blocks A, B, C; a row occupying two blocks is a side of the triangle, a pair is an angle (an endpoint in A on a row of type AC and one in B on a row of type BC, meeting at C), and no row fills the face, which is why (C3) never fires. Right: the same template in fixed width, rows grouped by type and columns by block; each type leaves one block empty, and a pair runs along a channel between two types through their common block (Lemma 4.1).

Why not four blocks, or two? With two blocks a row occupying both has no hole left; with k blocks of equal size a row occupies only $2/k$ of the columns, and three equal blocks balance occupation against holes. No configuration whatsoever does better than one third asymptotically: Lebedev's density bound [24, Thm. 3] gives $|E_2| \le mn/3 + O(m+n)$. For fixed n the error term is of the same order as the main term in m, so this bound does not give the slope; for $n \le 6$ the exact slope is λ_n, about one half below $n/3$, and finding it is the subject of Section 6.

How many free cells does a pair use? Each singleton row leaves $n-1$ free cells. For $n = 3, 4$ the slope is exactly $(n-1)/4$, as if every pair used four free cells, its two endpoints and two holes. From $n = 5$ on the blocks do better: Figure 4 has 20 pairs on $110 - 32 = 78$ free cells, more than one pair per four free cells, and in the modules of Section 6 the number of pairs per free cell, $\lambda_n/(n-1)$, tends to one third from below. At most one pair per three free cells is what holds in every case we computed (Question 3).

How to read Figure 4. Every pair is an angle of the triangle, and the three corners are used. Pairs 1–8 join an endpoint in block B of a row avoiding A with an endpoint in block A of a row avoiding B:

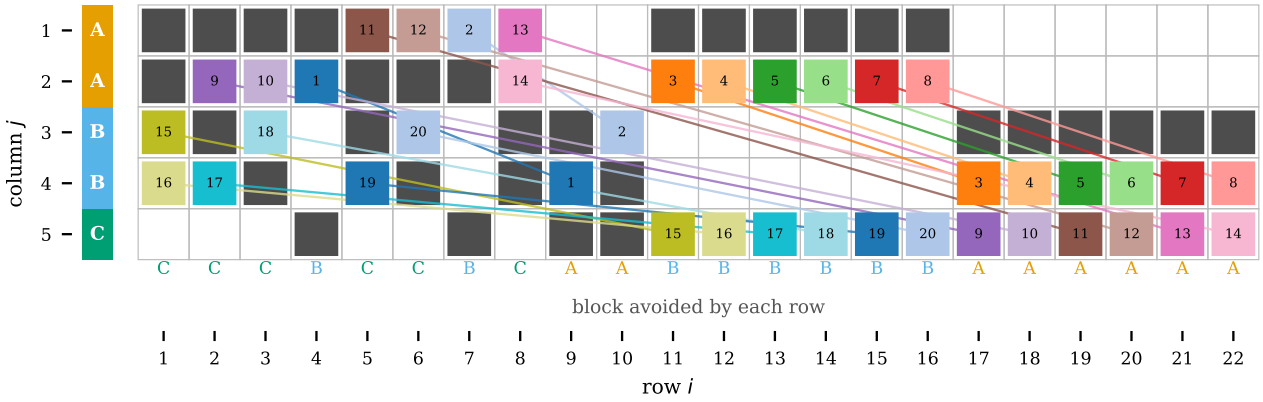


FIGURE 4. A three-block configuration on 22×5, drawn transposed (rows of the grid run horizontally). Blocks $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5\}$, the balanced sizes for $n = 5$. Rows 1–10 are the special rows (two grey cells each) and rows 11–22 singleton rows; the letter under each row is the block it avoids. The 32 cells of E_1 are grey and the 20 pairs are numbered in their two cells (16 endpoints lie in special rows, 24 in singleton rows). There are $110 - 32 = 78$ free cells and $20/78 > 1/4$ pairs per free cell; the 40 endpoints occupy more than half of the free cells. Data in `witness_22x5_three_blocks.json`, checked by both admissibility checkers.

eight angles at C. Pairs 9–14 join block C of a row avoiding A with block A of a row avoiding C: six angles at B. Pairs 15–20 join block C of a row avoiding B with block B of a row avoiding C: six angles at A. In each pair the two opposite cells fall in blocks that their rows avoid, so (C2) holds, and no row meets all three blocks, so no witness for (C3) exists. The special rows take part too: each avoids the block it does not touch.

5. THREE COLUMNS

Where do the holes come from when there are only three columns? For $m \geq 3$ a maximum base consists of the rows $\{1, 2\}, \{1, 3\}, \{2, 3\}$ and $m - 3$ rows with a single one. For a row r let S_r be its occupied support; the cells of S_r outside E_1 are its *endpoints*, and $2|E_2|$ is the number of endpoints. Two observations do all the work.

Lemma 5.1 (vertical pairs, any n)**.** *If $m \geq \binom{n}{2}$ and $\{(i, j), (p, j)\} \in E_2$, then $S_i \cap S_p = \{j\}$.*

Proof. If $\ell \neq j$ lies in both supports, the row k of E_1 through the pair of columns $\{j, \ell\}$ is distinct from i, p (its cell in column j is in E_1, theirs are not) and gives the forbidden cells $(k, \ell), (k, j), (i, \ell), (p, \ell)$ of (C3). □

Lemma 5.2 (one full row, any n)**.** *If $n \geq 3$ and $m \geq \binom{n}{2}$, at most one row is completely occupied.*

Proof. Let $F \neq G$ be full. No row of the base has more than two cells, so F has an endpoint (F, j). A partner (F, q) in the same row is excluded by G and any third column via (C3). A partner (P, j) in the same column contradicts Lemma 5.1, since S_P contains the E_1-cell of P, outside column j. A partner (P, q) with $P \neq F$, $q \neq j$ forces $(P, j) \notin O$ by (C2), as $(F, q) \in O$; so P is not full, $P \neq G$, an E_1-cell of P lies in a column $\ell \notin \{j, q\}$, and $(G, \ell), (G, j), (G, q), (F, \ell), (P, \ell) \in O$ violates (C3). □

Every other row keeps an empty cell, so a special row carries at most $n - 3$ endpoints, a singleton row at most $n - 2$, and the full row one more. Hence, for every $n \geq 3$ and $m \geq \binom{n}{2}$,

$$2F_n(m) \;\le\; (n-2)\,m - \binom{n}{2} + 1.$$

For $n = 3$ this is the exact bound of Theorem A; for larger n the slope $(n-2)/2$ is weaker than $\lambda_n \approx n/3$, and the real work is in Section 6. Lemmas 5.1 and 5.2 and this bound are formalized in Isabelle/HOL (see Formalizations at the end).

Proof of Theorem A. A row that is not full has at least one empty cell, so a singleton row carries at most one endpoint and a special row none, except the single full row of Lemma 5.2, which carries at most two. Hence $2|E_2| \le (m-3)+1 = m-2$ for every maximum base. Equality is reached: for m odd pair the singleton rows (all with their one in column 1) two by two as $\{(r,2),(r',3)\}$. For m even take one singleton row t with its one in column 2 and the pair $\{(s,1),(t,3)\}$, where s is the special row $\{2,3\}$, which becomes the full row; split the other $m-4$ singleton rows evenly between ones in columns 1 and 2, and join them by vertical pairs in column 3. For a vertical pair C is empty; for the other pairs $B \times C$ lies in column 1 (odd case) or column 2 (the last pair), where every row of B is empty, and (C2) holds because at least one opposite cell is free; so the families are admissible. □

The definitions of Section 2 and this proof are formalized in Lean 4 with Mathlib, with no axioms beyond the standard ones; the module is

`Formalizations.Combinatorics.Zarankiewicz.LimitedAugmentedThree`

in the public repository `https://github.com/karlesmarin/formalizations` (commit `389a576`, Lean `v4.35.0`, Mathlib `f0b4c81`; main theorem `ZL.zL_three`, checked with `lake build`). It proves the upper bound for every maximum base, not only for one, and builds the families above.

Theorem A also settles the second half of the open problem of [28], which asks whether the small gap they observed between the weak number $z_{wL}(m,3)$ and $z_L(m,3)$ persists. They prove $z_{wL}(m,3) = m+3+\lfloor(2m-4)/3\rfloor$ for $m \ge 16$ [28, Thm. 26], so

$$z_{wL}(m,3) - z_L(m,3) = \Big\lfloor \frac{2m-4}{3} \Big\rfloor - \Big\lfloor \frac{m-2}{2} \Big\rfloor = \frac{m}{6} + O(1) \qquad (m \ge 16)\,:$$

the gap grows linearly. The weak conditions allow slope $2/3$ per row, the original ones only $1/2$.

With this, five of the numbers studied in the series are known exactly in three columns: the original z_L, the weak z_{wL} of [27, 28], and the recursive-line, signed and second-order numbers z_{RL}, z_{SL}, z_2 of Löfberg and Qi [18] (relaxed versions of the irreducibility test behind z_L). They prove $z_2(m,3) = z_{RL}(m,3) = 2m$ for $m \ge 3$ [18, Prop. 4.13], and their chain $z_2 \ge z_{SL} \ge z_{RL} \ge z_{wL} \ge z$ [18, Prop. 7.4] then gives $z_{SL}(m,3) = 2m$ as well; it sits above $z_L \le z_{wL}$ [27]. (z_A and BSR in three columns are not determined here.) Measured from $z(m,3) = m+3$, each new row buys

$$\tfrac{1}{2} \text{ pair for } z_L, \qquad \tfrac{2}{3} \text{ for } z_{wL}\ (m \ge 16), \qquad 1 \text{ for } z_{RL} = z_{SL} = z_2,$$

and the three formulas $\lfloor 3m/2\rfloor + 2$, $m+3+\lfloor(2m-4)/3\rfloor$ and $2m$ are consistent with the published gap $z_2(m,3) - z_{wL}(m,3) = \lfloor(m-3)/3\rfloor$ for $m \ge 16$ [18, Cor. 4.14], which we rechecked.

Card A — three columns

What it says. $z_L(m,3) = m+3+\lfloor(m-2)/2\rfloor$ for all $m \ge 3$; previously known for $m \le 13$.
Why it holds. A maximum base covers every pair of columns, so the two rows of a vertical pair share only that column, and at most one row fills up; every other row keeps a hole, which caps the endpoints at $m-2$.
Status. Theorem; proof above; formalized in Lean (no `sorry`, standard axioms).
Data. Lean module `LimitedAugmentedThree`; values for $m \le 14$ in `table_zL.json`.

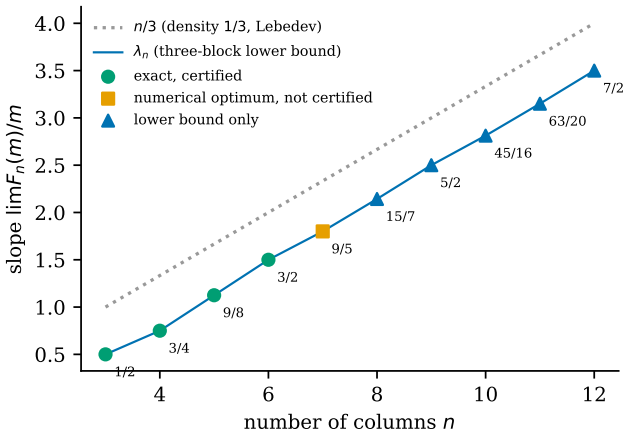


FIGURE 5. The slope $\rho_n = \lim F_n(m)/m$ for $n = 3, \dots, 12$. Green: $\rho_n = \lambda_n$ certified by rational duals ($1/2, 3/4, 9/8, 3/2$ for $n \le 6$); orange: the numerical candidate 1.8 for ρ_7 from a mixed integer program, compatible with $9/5$ but not certified; blue: the lower bound λ_n of the three-block modules (Theorem B), with ρ_n unknown ($n = 8$ was inconclusive). The solid line is λ_n and the dotted line the density $n/3$ of Lebedev's regime where both dimensions grow.

6. Slopes in fixed width

How many pairs does each new row buy when n is fixed? Lemma 5.1 already shows the price of a pair. The answer comes in two halves.

6.1. **Lower bound: modules around a hollow triangle.** Number the blocks $1, 2, 3$ and let a row of type $\{a, b\}$ occupy blocks a and b completely, apart from its single cell of E_1. For block sizes (B, c, c) a module consists of $2c-1$ rows of type $\{1, 2\}$ and $2c-1$ rows of type $\{1, 3\}$ with their cell of E_1 in block 1, $B-1$ rows of type $\{2, 3\}$ with it in block 2 and $B-1$ with it in block 3. For sizes (B, B, c) it consists of c rows of type $\{1, 2\}$ with the cell in block 1, c with it in block 2, $2B-1$ rows of type $\{1, 3\}$ with it in block 1 and $2B-1$ of type $\{2, 3\}$ with it in block 2. Matching the endpoints along the three channels of Lemma 4.1 (any bijection within a channel) gives

$$(B, c, c): \ (2c-1)(2B+c-2) \text{ pairs on } 2(2c+B-2) \text{ rows},$$
$$(B, B, c): \ (2B-1)(B+2c-1) \text{ pairs on } 2(2B+c-1) \text{ rows}$$

The special rows and any further singleton rows carry no endpoints and meet at most two blocks. With the module (B, c, c) of sizes (b, b, b) for $n = 3b$ and $(b+1, b, b)$ for $n = 3b+1$, and the module (B, B, c) of sizes $(b+1, b+1, b)$ for $n = 3b+2$, the ratio of pairs to rows is exactly λ_n; repeating the module gives $F_n(m) \ge \lambda_n m - O(1)$. Every module was rebuilt for $n = 3, \dots, 12$ and checked by two independent admissibility checkers (`lambda_modules.json`).

6.2. **Upper bound: shadows of triples.** Discard every pair meeting a special row or a row whose occupied support occurs fewer than three times; there are at most $(n-2)\binom{n}{2} + 2(n-1)(2^n - 1)$ of them, a constant. Each row keeps its original support as a *label*, even if discarding removes one of its endpoints; the inequalities below are capacities, so unused cells are allowed. The remaining pairs join singleton rows with *frequent* labels, and they obey:

- no pair lies inside one row (a second row with the same support gives a (C3) witness);
- a vertical pair in column j joins supports S, T with $S \cap T = \{j\}$ (Lemma 5.1);

- rows with the same support are never paired (by (C2), or by the previous rule for a vertical pair);
- a pair from (S, j) to (T, q), $j \neq q$, needs $q \notin S$ or $j \notin T$ (C2), and no $\ell \in (S \cap T) \setminus \{j, q\}$ with $\{j, q, \ell\}$ in the *shadow* $\mathcal{H}$, the set of triples of columns contained in some frequent support: such a triple lies in at least three rows, one of them external to the pair (C3).

In particular a full row carries no remaining pair: partners in its row or column are excluded by the first two rules, any other partner (T, q) would need $j \notin T$, and then the E_1-cell of T gives a forbidden ℓ. For each shadow this defines a linear program: variables $x_{S,a} \geq 0$, the proportion of rows with label S and cell of E_1 in column a (over supports all of whose triples lie in $\mathcal{H}$), with $\sum x_{S,a} \leq 1$, capacities $\sum_{a \neq j} x_{S,a}$ at the port (S, j), and flows on the permitted pairs of ports. Every actual configuration gives a feasible point, so the value of the program bounds $\limsup F_n(m)/m$. A dual solution, weights $y_{S,j} \geq 0$ with $y_{S,j} + y_{T,q} \geq 1$ on permitted pairs and a budget $\Lambda \geq \sum_{j \in S \setminus \{a\}} y_{S,j}$ for all (S, a), certifies the value $\leq \Lambda$: the flow is at most $\sum_{S,a} x_{S,a} \sum_{j \in S \setminus \{a\}} y_{S,j} \leq \Lambda$. The maximum over all shadows bounds the slope.

What does a certificate look like? For $n = 4$ there are five shadows up to symmetry, and only one, $\{123, 124\}$, reaches $3/4$: it is the three-block shadow with blocks $\{1, 2\}, \{3\}, \{4\}$. Its dual puts weight $1/4$ on the ports of the support $\{1, 2\}$, weight $3/4$ on the ports of the other two-column supports, and on the supports 123 and 124 weight $1/4$ on columns $1, 2$ and $1/2$ on the third column. Every permitted pair of ports carries total weight at least 1, and every row, whatever column holds its one, carries at most $3/4$; summing over rows gives at most $3m/4 + O(1)$ pairs. The other four shadows have certificates of value $1/2$ or $2/3$.

Proposition 6.1. *For $n = 3, 4, 5, 6$ the shadows form 2, 5, 34 and 2136 orbits under column permutations, and each orbit has a rational dual certificate of value at most λ_n; an independent checker in exact arithmetic verifies 33, 501, 14 316 and 2 892 596 inequalities.*

Together the two halves prove Theorem B. For $n = 7$ the 2^{35} shadows are too many to list; solving the shadow as 0/1 variables in one mixed integer program returns the numerical optimum 1.8, compatible with $9/5 = \lambda_7$; a decimal output does not certify the rational value, and we report it as evidence only.

6.3. **The slope always exists.** The linear programs do more than bound the slope: their optimum is attained.

Theorem C. *For every $n \geq 3$ let ρ_n be the largest value of the shadow programs over all shadows $\mathcal{H}$. Then ρ_n is rational and*

$$F_n(m) = \rho_n\, m + O_n(1) \qquad (m \to \infty).$$

Moreover $\rho_n \geq \lambda_n$, with equality for $n \leq 6$.

Proof. The upper bound $F_n(m) \leq \rho_n m + O_n(1)$ is the reduction above. For the lower bound take an optimal shadow and a rational optimal solution of its program, with $\sum x_{S,a} = 1$, and multiply by a common denominator D: this gives integers $X_{S,a}$ (rows with label S and E_1-cell in column a) and Y_e (pairs on each permitted pair e of ports). Build these rows as singleton rows; at the port (S, j) choose as many rows of label S with $a \neq j$ as the port needs, which the capacities allow, put an endpoint in column j of each, and join the endpoints along the pairs e. A row's real support is contained in its label. Two ports of the same label are never permitted, so the two cells of a pair lie in different rows. (C2) holds because supports only shrink. A vertical pair has $S \cap T = \{j\}$, so no column for a (C3) witness. For a diagonal pair, a witness row would contain three columns j, q, ℓ with $\ell \in (S \cap T) \setminus \{j, q\}$; the special rows are built with their two base cells only, so the witness is a labelled row, whose label puts $\{j, q, \ell\}$ in $\mathcal{H}$, which the permitted pairs exclude. The same argument applies to any number of copies of this module together with the $\binom{n}{2}$ special rows and filler singleton rows, so $F_n(m) \geq \rho_n D \lfloor (m - \binom{n}{2})/D \rfloor$. There are finitely many programs, all with rational data, so ρ_n is rational. The modules of Section 6.1 give $\rho_n \geq \lambda_n$, and Proposition 6.1 gives equality for $n \leq 6$. □

The admissibility step of this proof is a general *label lemma*: if every support lies in its label, any three occupied columns of a row form a triple of $\mathcal{H}$, and the pair is compatible with the labels, then (C2) and (C3) hold. It is formalized in Isabelle/HOL together with Lemma 4.1, which is the special case of labels made of two blocks (theory `ZL_Labels`, no `sorry`). We also checked the realization step by machine: for $n = 3, 4, 5, 6$ the program of the optimal shadow, built as above with two copies, gives configurations of sizes 7×3, 14×4, 26×5 and 31×6 with exactly ρ_n pairs per labelled row (1/2, 3/4, 9/8, 3/2), admissible under both of our checkers (`zl_rho_realiza.py`). So Question 1 is a question about one explicit finite family of linear programs.

How far is the original criterion from the relaxed ones? Lebedev has shown that the recursive-line, signed and second-order numbers all satisfy $z_X(m,n) = \frac{n+1}{2}m + O_n(1)$ for fixed n [15, Thm. 1.2]. Theorem C and the bound $\rho_n \le (n-2)/2$ that follows from Lemma 5.2 give at once:

Corollary 6.2. *For every fixed $n \ge 3$ and $X \in \{RL, SL, 2\}$,*

$$z_X(m,n) - z_L(m,n) = \Big(\frac{n-1}{2} - \rho_n\Big)m + O_n(1), \qquad \frac{n-1}{2} - \rho_n \;\ge\; \frac{1}{2}.$$

So the separation is linear in every fixed width; for $n = 3, 4, 5, 6$ the gap slope is 1/2, 3/4, 7/8 and 1. (In three and four columns the relaxed gain per row, $(n-1)/2$, happens to be exactly $2\rho_n$; from $n = 5$ on it is not.)

What is optimal inside the three-block family? Partition the columns into blocks A, B, C of sizes $x \ge y \ge z \ge 1$, $x + y + z = n$. An *ordinary module* is a finite set of R rows, each with exactly one base cell (its anchor), and P disjoint pairs of further cells, obeying the rule of Lemma 4.1: each row has a type of one or two blocks containing its anchor and its occupied support, with partial supports allowed. Let $M(x,y,z)$ be the supremum of P/R over ordinary modules. The maximum-base setting is recovered by adding the $\binom{n}{2}$ special rows bare, which costs a bounded number of rows; so M is a slope, not a ratio $F_n(m)/m$ (for instance $M(1,1,1) = 1/2$, while $F_3(m)/m < 1/2$ for every m).

Proposition 6.3. *If $z(x-1) \ge y(y-1)$, then $M = \frac{(s-1)((x-1)s+yz)}{s(n-2)}$ with $s = y + z$; otherwise $M = \frac{(u+1)(zu+(x-1)(y-1))}{u(n-1)}$ with $u = x + y - 2$. In the first case*

$$\frac{n}{3} - \frac{1}{2} - M \;=\; \frac{\eta(2\eta-1)}{6(n-2)} + \frac{(s-1)(y-z)^2}{4s(n-2)}, \qquad \eta = x - \frac{y+z}{2},$$

and in the second case

$$\frac{n}{3} - \frac{1}{2} - M \;=\; \frac{\eta(2\eta-1)}{6(n-1)} + \frac{(u+1)(x-y)^2}{4u(n-1)}, \qquad \eta = \frac{x+y}{2} - z.$$

(On the boundary $z(x-1) = y(y-1)$ the two formulas agree wherever both are defined; for $x = y = z = 1$ only the first is defined, and it gives $M = 1/2$.) Consequently $\max_{x+y+z=n} M = \lambda_n$, attained only by the sizes (k,k,k), $(k+1,k,k)$ and $(k+1,k+1,k)$ according as $n = 3k$, $3k+1$, $3k+2$.

Proof. Lower bound: modules in which every row fills its two blocks and pairs are matched along the three channels of Lemma 4.1. In the first case take $z(s-1)$ rows of type AB and $y(s-1)$ of type AC anchored in A, and $(x-1)y$, $(x-1)z$ rows of type BC anchored in B, C; the channels then carry $yz(s-1)$, $(x-1)z(s-1)$ and $(x-1)y(s-1)$ pairs, so $R = s(n-2)$ and $P = (s-1)((x-1)s+yz)$. In the second case take $z(y-1)$ and $z(x-1)$ rows of type AB anchored in A and B, $(y-1)(u+1)$ of type AC anchored in A, and $(x-1)(u+1)$ of type BC anchored in B; the channels carry $z(x-1)(u+1)$, $z(y-1)(u+1)$ and $(x-1)(y-1)(u+1)$ pairs, so $R = u(n-1)$ and $P = (u+1)(zu+(x-1)(y-1))$. Upper bound: weights on endpoint cells by row type and block, $(AB,A) \mapsto p$, $(AB,B) \mapsto q$, $(AC,A) \mapsto r$, $(AC,C) \mapsto 1-q$, $(BC,B) \mapsto 1-r$, $(BC,C) \mapsto 1-p$, with $p = r = \frac{s(s-1)-yz}{s(n-2)}$, $q = \frac{z}{s}$ in the first case and $p = q = \frac{zu+(x-1)(y-1)}{u(n-1)}$, $r = \frac{y-1}{u}$ in the second, and on one-block rows $w_A = \max(p,r)$,

$w_B = \max(q, 1-r)$, $w_C = \max(1-p, 1-q)$. Every pair of the block rule has weight at least 1; a row of any type and anchor carries at most M (the regime condition is exactly the inequality that makes the two non-active row costs at most M); and leaving cells unused only lowers a row's weight. Summing gives $P \le MR$. The identities are algebra, and η is a non-negative half-integer, so both terms are non-negative. For $n = 3k$ they vanish only at $x = y = z$; for $n = 3k+1$ the least deficit $1/(6(n-2))$ is attained only at $(k+1, k, k)$, in the first case, and for $n = 3k+2$ the least deficit $1/(6(n-1))$ only at $(k+1, k+1, k)$, in the second; the other case is excluded there by its regime condition. □

We checked the identities symbolically (`v189_identidades.py`) and, in exact arithmetic, for every partition with $n \le 60$: that every permitted pair has weight at least 1, that no row of any type and anchor exceeds M, that the weights lie in $[0, 1]$, that the modules balance and give $P/R = M$, and that the maximum is λ_n, attained only by balanced sizes (`tres_bloques_dual_check.py`). The proposition does not settle Question 1: a general configuration need not obey the block rule. It does explain the shape of the formula: writing $\lambda_n = \frac{n}{3} - \frac{1}{2} - \delta_n$, with $\delta_n = 0$, $\frac{1}{6(n-2)}$ or $\frac{1}{6(n-1)}$ according as $n \equiv 0, 1, 2 \pmod 3$, the term $-\frac{1}{2}$ is the price of reserving one base cell per row before halving the remaining capacity, and δ_n is the price of unequal integer blocks.

Does the base matter? In the regime $m \ge \binom{n}{2}$ all maximum bases have the same shape, but the singleton rows may put their one in any column. Lebedev shows that the *canonical* base, with all these ones in column 1, loses density: for $m \ge n^2$ it admits at most $mn/3 - c_0 n(m - n^2) + 4m$ pairs, for an absolute constant $c_0 > 0$ [24, Thm. 13], whereas, in the different regime $\min\{m, n\} \to \infty$, every maximum base admits $(1/4 - o(1))mn$ pairs [24, Thm. 9] (for fixed n this does not apply: $F_3(m) = m/2 + O(1) < 3m/4$). The realization above keeps the column of each E_1-cell, so restricting the programs to $a = 1$ gives the exact slope of the canonical base. We find $1/2$, $2/3$, 1 and $6/5$ for $n = 3, 4, 5, 6$, against $\rho_n = 1/2, 3/4, 9/8, 3/2$. Both directions are certified: rational dual certificates for every orbit of shadows under the permutations fixing the anchor column (2, 8, 90 and 10 688 orbits), checked by an independent program in exact arithmetic (`cert_canonica_check.py`), and realizations of the optimum, all anchored in one column, admissible under both of our checkers (`zl_rho_realiza.py -canonica`); for $n \le 5$ the values were also obtained by exact linear programming in Sage. For $n \ge 4$ the three-block modules of Section 6.1 need the ones of the base spread over the blocks, which is why the canonical base loses; for $n = 3$ both slopes are $1/2$. For $n = 4$ the canonical slope $2/3$ is the slope of the first branch of F_4 (Theorem E). All four values fit one formula, which holds for every width.

Theorem D. *For every $n \ge 3$ the largest number of pairs on the canonical base is $\kappa_n m + O_n(1)$, where*

$$\kappa_n = \frac{\lfloor (n-1)^2/4 \rfloor}{n-1},$$

the number of edges of a balanced complete bipartite graph on the $n-1$ free columns divided by $n-1$.

Proof. Let column 1 be the anchor, V the set of the $k = n-1$ free columns, and, for an ordinary row r of an arbitrary admissible configuration on the canonical base, $L_r = S_r \setminus \{1\}$ its *label*. A label is *frequent* if at least three rows carry it; at most $2 \cdot 2^k$ rows carry a rare label. Discard the pairs meeting a special row or a row with a rare label, at most $(n-2)\binom{n}{2} + 2(n-1)2^k$ of them. Let $\{(r, j), (r', q)\}$ be a remaining pair; its cells avoid column 1, which holds the base cells. It is not horizontal: if $r = r'$, a third row r'' with label L_r gives the (C3) witness $(r'', 1), (r'', j), (r'', q), (r, 1)$. It is not vertical: both rows contain column 1, against Lemma 5.1. So $r \ne r'$, $j \ne q$, and no frequent label contains both j and q, since among its three rows one, $r'' \notin \{r, r'\}$, would give the witness $(r'', 1), (r'', j), (r'', q), (r, 1), (r', 1)$. Let G be the graph on V whose edges are the pairs $\{j, q\}$ used by remaining pairs: every frequent label is an independent set of G. For any $p \in Q(G) = \{p \in \mathbb{R}^V_{\ge 0} : \ p_u + p_v \ge 1 \text{ for } uv \in E(G)\}$, charging each

pair $p_j + p_q \geq 1$ and each row the weights of its endpoints, which lie in its label, gives

$$F \;\leq\; \sum_r p(L_r) + O_n(1) \;\leq\; \gamma(G)\, m + O_n(1), \qquad \gamma(G) = \min_{p \in Q(G)} \; \max_{I \text{ independent}} p(I),$$

the critical threshold of the graphical simple game of G [16].

It remains to show $\gamma(G) \leq \lfloor k^2/4 \rfloor / k$ for every graph on k vertices; we follow the minimal-norm argument of [16, Thm. 3]. Let p^* be the point of minimal Euclidean norm of the closed convex set $Q(G)$, so $\langle p^*, q - p^* \rangle \geq 0$ for every $q \in Q(G)$. For an independent set I the indicator of $V \setminus I$ lies in $Q(G)$, which gives

$$p^*(I) \;\leq\; \sum_v p^*_v - \|p^*\|^2 \;=\; \frac{k}{4} - \left\| p^* - \tfrac{1}{2}\mathbf{1} \right\|^2 .$$

For even k this is the claim. Let k be odd, and call an edge *tight* if $p^*_u + p^*_v = 1$. A vertex with $p^*_v > 0$ on no tight edge could be lowered, so the vertices isolated in the tight graph have $p^*_v = 0$ and contribute $\frac{1}{4}$ to $\|p^* - \frac{1}{2}\mathbf{1}\|^2$. On a component of the tight graph with at least two vertices p^* alternates between t and $1 - t$ along tight edges. If the component is not bipartite, $t = \frac{1}{2}$. If it is bipartite with parts of sizes a, b, moving t to $t + \varepsilon$ on one part and $1 - t - \varepsilon$ on the other keeps p^* in $Q(G)$ for small ε of the admissible sign, so minimality forces $0 < t < 1$ and $at = b(1 - t)$, that is $t = b/(a + b)$; the component contributes $(a - b)^2/(4(a + b))$, which is at least $1/(4k)$ when $a \neq b$. Hence either $p^* = \frac{1}{2}\mathbf{1}$, and then $|I| \leq k/2$, so $|I| \leq \frac{k-1}{2}$ and $p^*(I) \leq \frac{k-1}{4}$; or $\|p^* - \frac{1}{2}\mathbf{1}\|^2 \geq \frac{1}{4k}$ and $p^*(I) \leq \frac{k}{4} - \frac{1}{4k}$. In both cases $\gamma(G) \leq \frac{k^2-1}{4k} = \lfloor k^2/4 \rfloor / k$.

For the lower bound split $V = X \cup Y$ with $|X| = \lceil k/2 \rceil$, $|Y| = \lfloor k/2 \rfloor$. A module has $|Y|$ singleton rows occupying X and $|X|$ singleton rows occupying Y, all anchored in column 1, and joins the $|X||Y|$ endpoints of each kind by any bijection: $\lfloor k^2/4 \rfloor$ pairs on k rows. A pair joins (r, x) with (r', y), $x \in X$, $y \in Y$; the opposite cell (r, y) is empty, which gives (C2); the two rows share only column 1, and no row contains 1, x and y (special rows have two cells), which gives (C3). Repeating the module and adding the special rows and filler rows gives $\kappa_n m - O_n(1)$ pairs. □

The formula agrees with the certified values for $n \leq 6$. Freixas and Kurz had already met the quantity $\lfloor k^2/4 \rfloor / k$ as an attained lower bound and conjectured maximum for general simple games, with a parameter normalized differently [10]; the proof above covers only graphical games, which is what the canonical base needs, and does not settle their conjecture.

Card B — slopes in fixed width

What it says. For every n each new row buys ρ_n pairs, a rational number given by finitely many linear programs, up to a bounded error; $\rho_n \geq \lambda_n = n/3 - 1/2 + O(1/n)$, with equality for $n \leq 6$.
Why it holds. Lower bound: modules around a hollow triangle (Lemma 4.1). Upper bound: after discarding boundedly many pairs, a linear program per shadow of triples; its optimum is realized by actual configurations (Theorem C); for $n \leq 6$ rational duals certify $\rho_n = \lambda_n$.
Status. Existence and rationality of ρ_n and $\rho_n \geq \lambda_n$: theorems for every n; $\rho_n = \lambda_n$: certified for $n \leq 6$, a conjecture for $n \geq 7$, with the numerical candidate 1.8 for $n = 7$.
Data. `lambda_modules.json`; `shadow_certificate_n3..6.json` with an independent checker.

7. Exact values

How far can one compute? Exact values have been part of the Zarankiewicz problem from the start: Guy's tables of the 1960s [12, 13] were computed largely by hand, and their extension near the triple-system threshold is recent [3]. For small grids we enumerate the maximum bases up to isomorphism with nauty [19] and, for each, maximize $|E_2|$ by constraint programming. For $m \geq \binom{n}{2}$ the base has only special and singleton rows, so an integer program over the *numbers* of rows of each type, with integer flows between ports, gives $F_n(m)$ exactly: this reduces 14×4 from two hours to seconds and reaches $m = 40$. Figure 6 collects the values; fourteen are new.

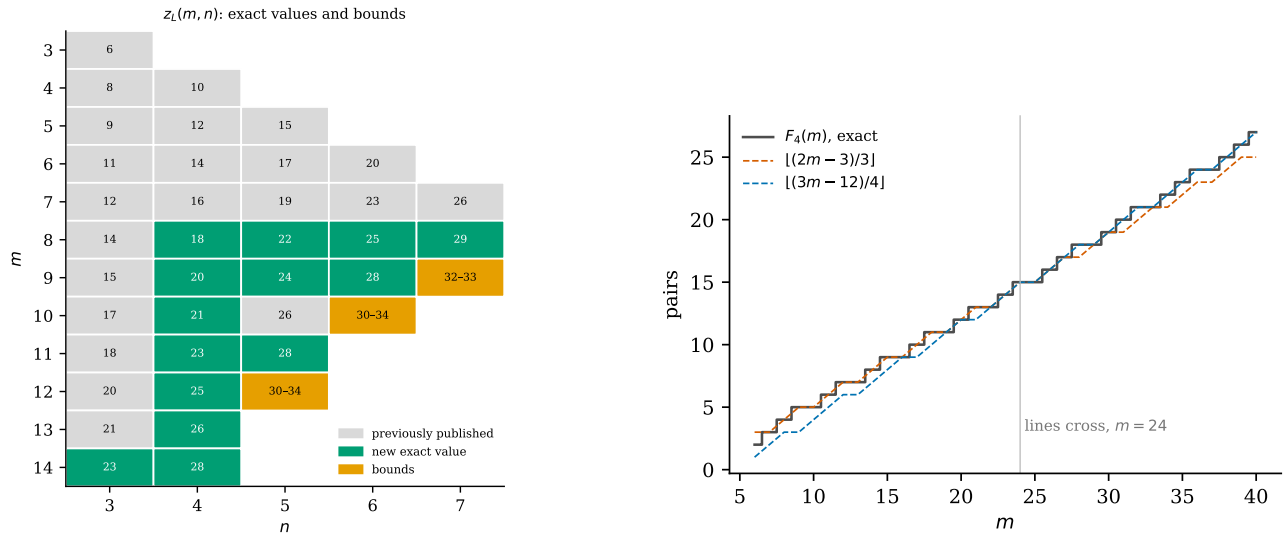


FIGURE 6. Left: the 39 values of $z_L(m,n)$ for $3 \le m \le 14$, $3 \le n \le 7$ that we computed, coloured by origin: 22 previously published (all reproduced), 14 new exact values, and 3 entries with lower and upper bounds (9×7, 10×6, 12×5). Right: $F_4(m) = z_L(m,4) - (m+6)$ for $6 \le m \le 40$ (staircase, from the type reduction) against the two floors $\lfloor (2m-3)/3 \rfloor$ and $\lfloor (3m-12)/4 \rfloor$; the lines cross at $m = 24$, and the second branch has the slope $3/4 = \rho_4$. Data in `table_zL.json` and `F4_exact.json`.

Can four columns be settled for every $m \ge 6$? Yes, by a finite description that is exact. For a configuration in the regime $m \ge \binom{n}{2}$ record, for every triple U of columns, its *profile* $p_U = \min(2, c_U)$, where c_U is the number of rows, special rows included, whose occupied support (base cells and endpoints) contains U. Then admissibility depends only on the profile and on the supports of the two rows of each pair: a vertical pair in column j is admissible iff $S \cap T = \{j\}$; a horizontal pair $\{j,q\}$ in a row of support S iff $p_{\{j,q,\ell\}} = 1$ for every $\ell \in S \setminus \{j,q\}$; a diagonal pair from (S,j) to (T,q) iff (C2) holds and, for every $\ell \in (S \cap T) \setminus \{j,q\}$, the triple $U = \{j,q,\ell\}$ lies in fewer than $1 + [U \subseteq S] + [U \subseteq T]$ rows (so that no row outside the pair contains it). Indeed a (C3) witness for a vertical pair is any further row occupied in j and in a common column ℓ, and the special row of $\{j,\ell\}$ is always one; for a horizontal or diagonal pair it is a further row whose support contains j, q and ℓ, which is counted in c_U whatever its base; and (C2) forbids both rows of a diagonal pair to contain U. For a fixed profile, the numbers of rows of each type and of pairs between each two ports satisfy a finite system of linear equations and inequalities, written out in Appendix A together with the realization of its integer solutions.

Theorem E. *For every $m \ge 7$,*

$$F_4(m) = \max\left\{ \left\lfloor \frac{2m-3}{3} \right\rfloor, \ \left\lfloor \frac{3m-12}{4} \right\rfloor \right\},$$

and $F_4(6) = 2$. In particular $z_L(m,4) = m + 6 + F_4(m)$. The first floor is at least the second for $m \le 27$ and the second, of slope $3/4 = \lambda_4$, is at least the first for $m \ge 22$; the lines cross at $m = 24$.

Proof (computer-assisted). The four triples give $3^4 = 81$ profiles, 15 up to permuting the columns; a displayed nondecreasing tuple such as $(0,0,2,2)$ denotes an orbit, its positions not tied to a fixed order of the triples (for $(0,0,2,2)$ we rename the columns so that the triples of profile 2 are 123 and 124). For each, the linear relaxation of the system above, with the valid inequality "at most one full row" (Lemma 5.2), has a rational dual certificate of $4f - 3m \le -12$ for the profile $(0,0,2,2)$ and of

$3f - 2m \le -3$ for the other fourteen, where f is the number of pairs. The duals were computed in exact arithmetic (Sage) and checked by an independent program that rebuilds the types, ports and permitted pairs from scratch and verifies every column (29 certificates: the 15 orbits for $4f - 3m$, of which only $(0, 0, 2, 2)$ is used, and 14 for $3f - 2m$; `cert_perfiles_check.py`). This gives the upper bound for every $m \ge 6$. For the lower bound, two explicit families, built from a seed by repeating a module of three rows and two pairs (slope 2/3, from $m = 7$) or of four rows and three pairs (slope 3/4, from $m = 10$), attain the two expressions; we checked them with both of our admissibility checkers for every $m \le 200$, and repeating the module adds rows whose supports contain only triples already of profile 2, so the profile stays the same in each residue class and, by the exact criterion, the repeated families stay admissible for every m. The case $m = 6$ is in the table. □

The programs are `zl_perfiles.py` (the system), `zl_perfiles_dual.sage` (the exact duals) and `f4_familias_check.py` (the families). The two branches have a partial structural reading. In the profile $(1, 1, 2, 2)$, which gives the slope-2/3 branch, a triple that occurs in a single row can serve as a (C3) witness for a whole channel of pairs and so blocks it; the fourteen profiles other than $(0, 0, 2, 2)$ each receive their own dual certificate of slope at most 2/3 (some of them, such as $(0, 0, 0, 0)$ or $(2, 2, 2, 2)$, contain no singly occurring triple at all). The profile $(0, 0, 2, 2)$ is the three-block shadow with blocks $\{1, 2\}, \{3\}, \{4\}$, and the intercept -3 is the cost of the six special rows in its dual. For comparison, Chen and Chen proved $z_{RL}(m, 4) = \lfloor (5m + 6)/2 \rfloor$ for $m \ge 15$ [4], so the gap to the recursive-line number in four columns is exactly $\lfloor (5m + 6)/2 \rfloor - m - 6 - F_4(m)$, of slope 3/4 (Corollary 6.2).

The same exact description gives eventual periodicity in every width.

Theorem F. *For every $n \ge 3$ there are integers $P_n \ge 1$ and M_n such that $F_n(m + P_n) = F_n(m) + \rho_n P_n$ for all $m \ge M_n$. If $\rho_n = a/b$ in lowest terms, every such P_n is a multiple of b.*

Proof. By Appendix A, the set of pairs (m, f) such that some admissible configuration with a maximum base on an $m \times n$ grid, $m \ge \binom{n}{2}$, has f pairs is a finite union, over the profiles, of projections of sets of non-negative integer solutions of linear systems with constant coefficients; so it is definable in Presburger arithmetic, and so is the graph of its maximum F_n (add that no realizable $g > f$ exists). By Theorem C, $D(m) = bF_n(m) - am$ is an integer function taking finitely many values. Each set $\{m : D(m) = k\}$ is definable in Presburger arithmetic, hence semilinear [11], hence eventually periodic as a subset of $\mathbb{N}$. A common period P and threshold for these finitely many sets give $D(m + P) = D(m)$, which is the claim with $P_n = P$; since F_n is an integer, $\rho_n P_n$ must be one, so $b \mid P_n$. □

Theorem E is the case $n = 4$ made explicit, with minimal period 4 and threshold 22.

Card C — exact values

What it says. Fourteen new exact values (widths 3 to 7, $m \le 14$), bounds for three more entries, and $z_L(m, 4)$ for every $m \ge 6$ (Theorem E), with two linear branches of slopes 2/3 and 3/4.
Why it holds. Small grids: exhaustive search over all maximum bases up to isomorphism and an exact integer program over row types, which reproduce every published value and agree where both apply. Four columns: exact profile system, rational dual certificates checked independently, and two periodic families checked by both checkers.
Status. Computer-assisted exact values; F_4 is certified for every $m \ge 6$ (and the regime $m \ge \binom{4}{2}$ starts at 6).
Data. `table_zL.json`, `F4_exact.json`, `perfiles_n4/`; programs in the archive.

8. One dimension more

If pairs live around a hollow triangle, do they live around a hollow tetrahedron one dimension up? The Zarankiewicz problem itself was lifted to higher dimensions early: Erdős showed in 1964 that an $n \times n \times n$ array of zeros and ones with no box of ones has $O(n^{3-1/4})$ ones [8]. Replace the matrix by an $m \times n \times p$ array, rectangles by boxes $\{a, a'\} \times \{b, b'\} \times \{c, c'\}$, and the base by a maximum box-free set. Two cells of O are adjacent when they are opposite corners of a box whose eight cells lie in O, and the

two cells of a pair must be at distance at least three in this graph: the distance form of (C2)–(C3) from Section 2. In the plane every computed case has at most one pair per three free cells, and three blocks are a triangle; a hollow tetrahedron of four blocks suggests at most one pair per four free cells here, $4|E_2| \le mnp - |E_1|$. We take all sides at least three, and the guess fails already for $3 \times 3 \times 5$.

Proposition 8.1. *In $3 \times 3 \times 5$ a maximum box-free base has 34 cells, and there is an admissible family of 3 pairs, so that $4|E_2| = 12 > 11$ free cells.*

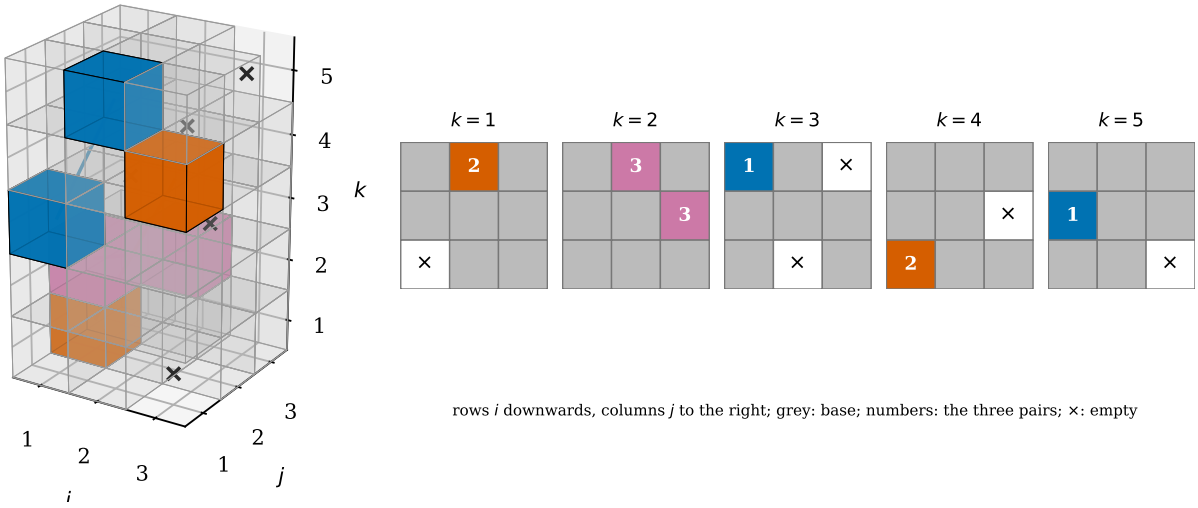


FIGURE 7. The $3 \times 3 \times 5$ configuration of Proposition 8.1. Left: in perspective. Right: the five 3×3 slices $k = 1, \dots, 5$, where every cell can be checked: the 34 base cells grey, the two cells of each of the 3 pairs with its number, and the 5 empty cells ×. The 11 free cells are 6 paired and 5 empty, so $4|E_2| = 12 > 11$. Data in `witness_3x3x5_tensor.json`.

The base is maximum because a 3×3 slice with a ones contains at least $2a - 12$ full rectangles and two slices of a box-free array never share one, so $2|E_1| - 60 \le 9$. The configuration is shown in Figure 7 and listed in `witness_3x3x5_tensor.json`; we checked it with our own box-graph checker. It is the smallest example by volume with all sides at least three: the same counting shows that every maximum base of $3 \times 3 \times p$, $p \le 5$, consists of slices with 6, 7 or 8 ones and pairwise disjoint sets of full rectangles, which leaves 9, 162 and 1413 bases up to permuting slices for $p = 3, 4, 5$, and an exhaustive search over them gives at most 0 pairs in $3 \times 3 \times 3$ and 2 pairs in $3 \times 3 \times 4$, where $4|E_2|$ only equals the number of free cells. Whether this variant has a meaning for sums of squares of trilinear forms is open.

Card D — one dimension more

What it says. The hollow-tetrahedron guess of one quarter fails already in $3 \times 3 \times 5$, the smallest volume where it can fail with all sides at least three.
Why it holds. An explicit configuration with 3 pairs on 11 free cells, admissible in the box graph; maximality of the base by counting full rectangles in the 3×3 slices; exhaustive search over the maximum bases of $3 \times 3 \times 3$ and $3 \times 3 \times 4$.
Status. Proposition, checked by computer; the meaning of the variant for trilinear forms is open.
Data. `witness_3x3x5_tensor.json`.

9. WHAT EACH STATEMENT RESTS ON

Table 1 lists every statement with its level of guarantee, its basis and the file where it can be checked. The levels, coloured as in Figure 1, are: *formal* (proof checked by Lean or Isabelle/HOL), *certified*

(finite certificate checked in exact arithmetic by an independent program), *computer* (exhaustive search or witness checked by our programs), *proof* (proof in the text), *evidence* (numerical only), and *partial* or *open*.

Table 1: What each statement rests on.

Level	Statement	Basis	Where to check
formal	Label, blocks, vertical, one full row	proofs in the text; Isabelle/HOL, from the properties of a maximum base (Formalizations)	theory `ZL_Labels`
formal	Distance form of (C2)–(C3); endpoint bound	proofs in the text; Isabelle/HOL	theory `ZL_Counts`
formal	Theorem A	proof in §5; Lean	module `LimitedAugmentedThree`
proof	Weak gap $m/6 + O(1)$	Theorem A and the cited formula [28, Thm. 26]; the floor arithmetic is formal	theory `ZL_Counts` (`weak_gap_arith`)
proof	Three-column ladder $\frac{1}{2}, \frac{2}{3}, 1$	Theorem A, [28], [18]	`cruce_escalera_m3.py`
proof	λ_n lower bound, all n	explicit modules; Lemma 4.1 (formal)	`lambda_modules.json`, two checkers ($n \leq 12$)
certified	Slopes exact, $n \leq 6$	shadow reduction + rational dual certificates	`shadow_certificate_n3..6.json`, independent checker
evidence	Slope $9/5$ for $n = 7$	mixed integer program, floating point	not certified
proof	Slope ρ_n exists, rational	Theorem C: reduction + realization	`zl_rho_realiza.py` ($n \leq 6$, two checkers)
proof	Linear gap to z_{RL}, z_{SL}, z_2	Corollary 6.2: Theorem C, Lemma 5.2, [15]	proof in the text
proof	Balanced blocks optimal in the family (Prop. 6.3)	explicit modules and dual weights; checked exactly for $n \leq 60$	`tres_bloques_dual_check.py`, `v189_identidades.py`
certified	Canonical slopes, $n \leq 6$	rational dual certificates + realizations	`cert_canonica_3..6.json`, independent checker
proof	Canonical slope, all n (Theorem D)	graph reduction + minimal-norm bound [16]; agrees with the certified $n \leq 6$	proof in the text
computer	New exact values	nauty classes + CP-SAT; type reduction	`table_zL.json`
certified	$F_4(m)$, every $m \geq 6$ (Theorem E)	profile system + rational duals; periodic families	`perfiles_n4/`, `cert_perfiles_check.py`; two checkers ($m \leq 200$)
computer	Proposition 8.1	explicit witness, maximality by counting	`witness_3x3x5_tensor.json`, box-graph checker
computer	$3 \times 3 \times 5$ smallest by volume	exhaustive search over the maximum bases of $3 \times 3 \times 3$, $3 \times 3 \times 4$	search programs in the archive
proof	Exact profile system (Appendix A)	both directions proved in the text	`zl_perfiles.py`
proof	Eventual periodicity, every n (Theorem F)	Theorem C, Appendix A, Presburger arithmetic [11]	proof in the text

10. Open questions

Question 1. Is $\rho_n = \lambda_n$ for every n? Equivalently (Theorem C), do three-block modules give $F_n(m) - O_n(1)$ pairs as $m \to \infty$? For $n \leq 6$ the optimal shadows are three-block ones (certified); for $n = 7$ the numerical candidate 1.8 equals λ_7. The question is about slopes, not exact optima: no exactly optimal configuration obeys the block rule of Lemma 4.1 when $n = 3$ and m is even. Indeed, then $2F_3(m) = m - 2$, while without a full row the special rows carry no endpoints and the other rows at most one each, so $2F_3(m) \leq m - 3$; hence every optimum has a full row carrying an endpoint, and under the rule a full row carries none, since its type contains every block. Optimal configurations are also not all close to the rule: the even construction of Theorem A uses $(m-4)/2$ vertical pairs, which the rule never allows, and the optimal four-column families of Theorem E repeat a module with a pair whose

opposite corner is occupied, so in both cases a linear number of rows must be removed to reach the rule. Separate questions are the stability of the *supports* or of the shadow (are the occupied supports of almost every row unions of two of three blocks?) and the classification of all optimal pairings. A positive answer to the first question would fix the gap slope of Corollary 6.2 at $g_n = n/6 + \delta_n$; the relaxed criteria would then buy $(n-1)/(2\lambda_n)$ times as many pairs per row as the original one, which is 2 for $n = 3, 4$, 16/9 for $n = 5$, 5/3 for $n = 6$, and tends to 3/2 only as $n \to \infty$.

Question 2. Theorem F says that $F_n(m) - \rho_n m$ is eventually periodic for every n, and Theorem E gives period 4 and threshold 22 for $n = 4$. What are the minimal period and threshold, and the explicit formula, for $n = 5, 6$? The denominators force the period to be a multiple of 8 and 2; the profile certificates of Section 7 could settle them, as they did for $n = 4$.

Question 3. Is there an exact inequality $3|E_2| + |E_1| \le mn$ for every maximum base, without the error terms of the density bound? It fails if the base is not maximum, and it holds in every case we computed. For $n = 3, 4$ and $m \ge \binom{n}{2}$ it follows from the bound $2F_n(m) \le (n-2)m - \binom{n}{2} + 1$ of Section 5; from $n = 5$ on that bound is too weak (it allows a violation from $m = 11$ for $n = 5$). Asymptotically the inequality says $\rho_n \le (n-1)/3$, which holds for $n \le 6$ by the certified slopes ($9/8 < 4/3$, $3/2 < 5/3$) and would hold for every n if $\rho_n = \lambda_n$, since $\lambda_n < (n-1)/3$; the exact inequality for every m is open from $n = 5$ on.

Question 4. What replaces the three blocks when $m < \binom{n}{2}$, where maximum bases are linear hypergraphs with rows of several sizes rather than pairs of columns, for instance when the base is a projective plane?

Question 5. Which three-dimensional analogue, if any, bounds the sum-of-squares rank of trilinear forms, and what plays the role of the hollow triangle there?

Question 6. The canonical base (all singleton ones in one column) has slope $\lfloor (n-1)^2/4 \rfloor/(n-1)$ (Theorem D), and the best bases reach ρ_n. What is the slope for a prescribed distribution of the singleton ones over the columns? Adding these frequencies to the shadow programs gives a parametric family of linear programs that contains both the canonical and the optimal distributions; is the slope a concave function of the distribution?

Appendix A. The exact profile system

Fix $n \ge 3$, $m \ge \binom{n}{2}$ and a profile $p \in \{0,1,2\}^{\binom{[n]}{3}}$. A *type* is a triple $\tau = (B, S, M)$ with $B \subseteq S \subseteq [n]$, $|B| \in \{1, 2\}$, no triple of profile 0 inside S, and M a partial matching of $S \setminus B$ (the horizontal pairs of the row) such that $p_{\{j,q,\ell\}} = 1$ for every $\{j, q\} \in M$ and $\ell \in S \setminus \{j, q\}$. Its *ports* are the columns $J_\tau = S \setminus (B \cup V(M))$: cells that must be paired with another row. Two ports (τ, j), (τ', q) of different types with supports S, S' are *permitted* if $j = q$ and $S \cap S' = \{j\}$, or $j \ne q$, not both $q \in S$ and $j \in S'$, and every $U = \{j, q, \ell\}$ with $\ell \in (S \cap S') \setminus \{j, q\}$ has $p_U < 1 + [U \subseteq S] + [U \subseteq S']$.

Variables: $x_\tau \in \mathbb{Z}_{\ge 0}$ (rows of type τ) and $y_e \in \mathbb{Z}_{\ge 0}$ (pairs on each permitted pair e of ports). Constraints:

$$\sum_\tau x_\tau = m, \qquad \sum_{\tau:\, B_\tau = B} x_\tau = 1 \quad \Big(B \in \tbinom{[n]}{2}\Big),$$

$$c_U = \sum_{\tau:\, U \subseteq S_\tau} x_\tau : \quad c_U = 0,\ c_U = 1,\ c_U \ge 2 \ \text{ according as } p_U = 0, 1, 2,$$

$$\sum_{e \ni (\tau, j)} y_e = x_\tau \qquad (\tau,\ j \in J_\tau).$$

The objective is $f = \sum_\tau |M_\tau| x_\tau + \sum_e y_e$. The port equations are equalities: every cell of $S \setminus B$ announced in a support is used by exactly one pair, so the realized supports are exactly the announced ones and the counts c_U are the true ones. (In Section 6 capacities sufficed, because supports there are frozen labels and real supports may be smaller.)

From configurations to solutions. Recording for every row its base, its occupied support and its horizontal pairs, and counting the other pairs between ports, gives a solution for the profile of the configuration: the constraints are counts, and every pair is permitted by the exact criterion of Section 7.

From solutions to configurations. Given an integer solution, create x_τ rows of each type, with base B_τ and the horizontal pairs M_τ. For each port (τ, j), the equation lets us assign the pairs of the incident edges to distinct rows of type τ, using column j of each; along each edge, match the two lists. The two cells of a pair lie in rows of different types, hence in different rows. Every non-base cell of every support is used exactly once, so the supports and the counts c_U are as prescribed, the profile is p, and the exact criterion makes every pair admissible. The rows with $|B| = 2$ give one row per pair of columns and the others one cell each, so the base is maximum. Hence the set of pairs (m, f) realized by admissible configurations is the union over the profiles of the projections of the solution sets, as used in Theorems E and F; for Theorem E the linear relaxation is strengthened by the valid inequality "at most one full row" (Lemma 5.2).

Formalizations

Lean 4. Theorem A with the definitions of Section 2: module `Formalizations.Combinatorics.Zarankiewicz.LimitedAugmentedThree` of the repository https://github.com/karlesmarin/formalizations, commit `389a576`, Lean `v4.35.0`, Mathlib `f0b4c81`; standard axioms only, checked by the repository's continuous integration.

Isabelle/HOL. Directory `Isabelle/AFP/Limited_Augmented_Zarankiewicz` of the same repository, commit `5754a1f`, built with Isabelle2026 without `quick_and_dirty` and submitted to the Archive of Formal Proofs. Theory `ZL_Labels`: the label lemma (`label_admissible`), Lemma 4.1 (`blocks_admissible`), Lemma 5.1 (`vertical_pair`) and Lemma 5.2 (`at_most_one_full_row`). Theory `ZL_Counts`: the distance form of (C2)–(C3) (`C2_iff_not_adj`, `C3_iff_no_common_neighbour`), the bound $2F_n(m) \le (n-2)m - \binom{n}{2} + 1$ (`endpoint_bound`) and the floor arithmetic of the weak gap (`weak_gap_arith`). The two lemmas on maximum bases and the bound are proved inside a locale whose hypotheses are the properties of a maximum base used in the text: every pair of columns is covered by a base row, every occupied row has one or two base cells, all cells lie in the grid, and $|E_1| = m + \binom{n}{2}$.

Data and code

The supplementary archive deposited with this paper on Zenodo contains: the witnesses (`datos/`, JSON, indices from 1); the shadow certificates for $n \le 6$ (`shadow_certificate_n3..6.json`) with the generator and the independent checker (`zl_cert_sombras.py`, `cert_sombras_check.py`); the canonical certificates (`cert_canonica_3..6.json`, `zl_cert_canonica.py`, `cert_canonica_check.py`); the profile system for four columns (`zl_perfiles.py`, `zl_perfiles_dual.sage`, the 29 certificates and `cert_perfiles_check.py`) and the check of the periodic families (`f4_familias_check.py`); the realizations (`zl_rho_realiza.py`), the module builder, the type reduction, the exact linear programs in Sage, the three-block check (`tres_bloques_dual_check.py`), the two admissibility checkers, and the figure scripts, which re-check every witness before drawing it. The exact values of the table come from exhaustive search (nauty and CP-SAT reaching proven optimality), not from independently checked proof certificates.

Use of AI tools

AI assistants were used as research assistants throughout, for computation, writing and review. The author checked every argument as described in the text and takes full responsibility for the content; results that are only computed or only proposed are marked as such in Table 1.

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